

RegTrace-Agent: Learning Regulatory Review Policies from Context-to-Feedback Traces

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Abstract

Regulatory, audit, and compliance workflows often ask a harder question than document QA: did an organization’s claimed fix actually resolve the reviewer request in the revised evidence? We introduce RegTrace-Agent, a framework that converts such workflows into context-to-feedback traces. A domain constructor maps local documents into request, response, evidence, label, feedback, and external-corroboration roles; a reviewer retrieves evidence, checks obligations, emits a guarded verdict, and uses structured natural-language feedback to improve the next reviewer policy. We instantiate the framework on SEC comment-letter review, releasing RegTrace-SEC, 472 public EDGAR examples with amended-filing evidence, visible-evidence labels, structured feedback, and later same-topic SEC follow-up signals. We also test cross-regulatory adaptation on CMS-2567 Plans of Correction and a FDA warning-letter stress test. On SEC grouped, temporal, and topic-holdout evaluations, full natural-language feedback is the strongest reviewer-training signal, outperforming handwritten prompting, MIPROv2, scalar/category GEPA variants, supervised lexical and encoder scorers, and typed decision gates. On CMS, domain-specific full-feedback optimization improves a new corrective-plan reviewer; on FDA, the constructor correctly reshapes the task when public response letters are not observable. Evidence ablations, independent audit, follow-up corroboration, and prompt-evolution analysis suggest that feedback-rich regulatory traces can support reusable evidence-grounded review policies rather than one-off prompt tuning.

1 Introduction

Financial NLP has rapidly moved toward LLM systems for long documents, domain-specific reasoning, and decision support. Models and benchmarks such as BloombergGPT, FinGPT, FinBen,

FinanceBench, FinQA, TAT-QA, and ConvFinQA show both the promise of financial LLMs and the continuing difficulty of grounding answers in financial evidence (Wu et al., 2023; Yang et al., 2023; Xie et al., 2024; Islam et al., 2023; Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022). Recent FinNLP work has similarly emphasized retrieval-augmented financial QA, SEC filing QA, financial intelligence benchmarks, factuality judging, and evasive-answer detection (Lithgow-Serrano et al., 2025; Lai et al., 2025; Klimaszewski et al., 2025; Wu et al., 2025; Al Nuaimi et al., 2025). Most such settings, however, still present a static input and ask for a static answer, label, or report.

Many operational finance and compliance workflows have a different structure. A regulator, auditor, lender, or control function requests a correction, explanation, disclosure, or supporting artifact; the organization responds that it has revised an external document; the reviewer must decide whether visible evidence actually satisfies the request. For example, a company may reply that it revised a filing to address a requested forecast disclosure, while the amended filing only shows combined revenue projections and still omits the requested entity-level assumptions. The task is therefore not to judge whether the response sounds cooperative, but whether the repaired artifact visibly closes the original gap. We call this class of problems evidence-grounded response resolution. It is related to fact verification and document-level legal NLI (Thorne et al., 2018; Koreeda and Manning, 2021), but adds a prior reviewer request, a response that may only claim a revision, revised evidence that must be checked, and textual feedback about what remains missing.

Existing work leaves three limitations for this setting. First, benchmarks rarely ask whether an organization has repaired a documented deficiency. Second, response evaluation often has weak obligation grounding: a response can sound persuasive

085 while amended evidence omits a named disclosure,
086 quantification, exhibit, legal basis, or accounting
087 analysis. Third, prompt and pipeline optimization
088 often discards feedback structure by receiving only
089 scalar success, although review failures are natu-
090 rally described in text: what was requested, what
091 was claimed, what evidence showed, and what gap
092 remained.

093 We study these limitations through U.S. Securi-
094 ties and Exchange Commission (SEC) comment-
095 letter review. SEC staff issue comments request-
096 ing expanded disclosure, clarification, revised ac-
097 counting discussion, removal of unsupported state-
098 ments, or additional exhibits; companies respond
099 in writing, often with statements such as “the dis-
100 closure has been revised accordingly.” After review
101 completion, the SEC releases staff comment letters
102 and company responses through EDGAR. These
103 traces often point to amended filings and some-
104 times include later same-topic SEC follow-up com-
105 ments. Accounting research treats comment let-
106 ters as economically meaningful regulatory signals
107 (Cassell et al., 2013; Bozanic et al., 2017; Baugh
108 and Schmardebeck, 2023; Ryans, 2021); for NLP,
109 they provide observable request-response-evidence-
110 feedback traces.

111 We introduce RegTrace-Agent, a context-to-
112 feedback-review framework for evidence-grounded
113 reviewers. It reconstructs review traces, retrieves
114 revised or corrective context, decomposes requests
115 into obligations, adjudicates visible gaps, com-
116 piles scalar/category/full feedback, and optimizes
117 a guarded reviewer policy. The learned reviewer is
118 not a single hand-tuned prompt; it is the policy in-
119 side a system that gathers evidence, checks obliga-
120 tions, emits a guarded verdict, and uses adjudicated
121 failures to improve the next offline reviewer. This
122 framing is deliberately broader than SEC filings:
123 the same loop applies whenever an organization re-
124 sponds to a reviewer by claiming to fix a document,
125 process, artifact, or remediation plan.

126 This design makes response resolution a
127 strong testbed for language-feedback optimization.
128 Prompt-programming work treats LLM pipelines
129 as declarative programs whose prompts and demon-
130 strations can be optimized (Khatab et al., 2023);
131 GEPA argues that natural-language feedback can
132 expose reusable failure structure that scalar reward
133 discards (Agrawal et al., 2026). In our setting, that
134 structure is regulator-style: a mistake can be ex-
135 plained as a missing requested disclosure, unsup-
136 ported revision claim, absent exhibit, or incomplete

analysis.

137
The above limitations motivate three research 138
questions. RQ1 asks whether public review 139
traces can be converted into a benchmark for 140
evidence-grounded response resolution. RQ2 141
asks whether full natural-language feedback is 142
a stronger reviewer-training signal than scalar 143
prompt search and coarse categories. RQ3 asks 144
whether the resulting review policy generalizes 145
across threads, time, issue categories, and neigh- 146
boring regulatory domains. 147

We make four contributions. First, we de- 148
fine evidence-grounded response resolution and 149
introduce RegTrace-Agent, a context-to-feedback- 150
review framework with explicit modules for trace 151
construction, evidence retrieval, obligation check- 152
ing, feedback construction, reviewer optimization, 153
and guarded reuse. Second, we instantiate it as 154
RegTrace-SEC, a public EDGAR benchmark with 155
SEC comments, company responses, amended- 156
filing evidence, labels, feedback fields, split files, 157
model outputs, and release metadata. Third, we 158
show that full natural-language feedback is the 159
strongest reviewer-training signal, outperforming 160
handwritten prompting, MIPROv2, scalar/category 161
GEPA variants, supervised lexical and encoder 162
baselines, and typed decision gates under fixed 163
test-time evidence. Fourth, we validate the bench- 164
mark and mechanism through evidence ablations, 165
independent-feedback and feedback-component ab- 166
lations, prompt-evolution analysis, independent 167
LLM audit, systematic corroboration from later 168
SEC follow-up comments, and cross-regulatory 169
studies on CMS and FDA. These analyses sup- 170
port the central claim: the reusable object is not a 171
prompt string, but a trace-to-feedback method for 172
learning regulator-style review policies. 173

2 Related Work 174

Financial NLP and evidence-grounded evalua- 175
tion. Financial NLP has moved from broad docu- 176
ment classification toward long-document reason- 177
ing, evidence grounding, and compliance-sensitive 178
decisions. FinQA, TAT-QA, and ConvFinQA intro- 179
duced financial reasoning questions over reports, 180
tables, and conversations (Chen et al., 2021; Zhu 181
et al., 2021; Chen et al., 2022); FinanceBench eval- 182
uates open-book QA over company documents (Is- 183
lam et al., 2023). Recent FinNLP work has simi- 184
larly emphasized RAG over financial documents, 185
accessible finance-intelligence benchmarks, factu- 186

ality and causality judging, SEC filing QA, and evasive-answer detection (Lithgow-Serrano et al., 2025; Klimaszewski et al., 2025; Lai et al., 2025; Al Nuaimi et al., 2025; Wu et al., 2025). Our task differs by asking whether retrieved evidence satisfies a prior regulator request.

SEC comment letters and regulatory review. Accounting and disclosure research has studied SEC comment letters as regulatory signals, including remediation costs, disclosure effects, recurring disclosure deficiencies, and textual classification of comment-letter topics (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmarbeck, 2023; Ryans, 2021). That literature generally treats comment letters as economic or disclosure events. We instead treat the correspondence as document-grounded supervision for LLM systems. Each review unit contains a staff request, company response, amended filing evidence, and sometimes a next-round SEC follow-up.

Prompt optimization and language feedback. DSPy reframes LLM pipelines as declarative programs whose prompts and demonstrations can be optimized against metrics (Khatab et al., 2023). MIPROv2 and related optimizers primarily use scalar feedback. A broader line studies language feedback, including self-refinement, verbal reinforcement, and critique-based training (Scheurer et al., 2022; Madaan et al., 2023; Shinn et al., 2023). GEPA uses natural-language reflection over execution traces (Agrawal et al., 2026). This distinction is central here because SEC response errors are naturally textual: missing disclosures, unsupported revision claims, absent exhibits, incomplete analysis, or over-demand beyond the SEC request.

3 RegTrace-Agent Framework

3.1 General Review-Trace Setting

RegTrace-Agent applies to workflows where a response claims to satisfy a prior review request by modifying or supplying evidence outside the response itself. Formally, each review trace contains a request q , a response r , a collection of revised or supporting artifacts $\mathcal{D}$, and optional later reviewer feedback u^+ . The system retrieves evidence $e = R(q, r, \mathcal{D})$, extracts obligations $o_{1:m} = O(q)$, and applies a verifier $V(q, r, e, o_{1:m})$ whose output is either resolved or unresolved. The same trace can also produce feedback for learning. A scalar signal says whether the prediction is correct. A

category signal names a broad issue or gap type. A full natural-language signal describes the request, the response action, the visible evidence, and any unmet requirement. This makes the framework applicable beyond SEC review whenever request-response-evidence traces are available.

3.2 Adaptive Trace Construction

The first stage of RegTrace-Agent is a trace-construction agent rather than a fixed SEC preprocessor. For a candidate domain d , it emits a TRACESPEC

$$S_d = (\phi_d, y_d, f_d, a_d, \rho_d),$$

where ϕ_d maps local fields into request, response, evidence, metadata, and external-corroboration roles; y_d defines the review label; f_d defines scalar, category, and full-text feedback channels; a_d lists fields forbidden at test time; and ρ_d is a utility decision. The utility decision scores role observability, evidence availability, label viability, feedback richness, external corroboration, release viability, and optimization suitability. If any core role is weak, the agent must either reshape the task or reject it instead of forcing the SEC schema.

This matters because regulatory corpora expose different trace structures. In SEC review, the appropriate trace is staff comment, company response, amended-filing snippets, visible-evidence label, and later same-topic follow-up. In CMS-2567, the response and corrective artifact are often the same Plan of Correction, so the task becomes plan adequacy rather than amended-document verification. In a probe over FDA warning letters, public warning and closeout letters are available, but public company response letters are largely absent; the construction decision is therefore to reshape the domain into warning-to-closeout resolution tracing or add contrastive controls before treating it as a benchmark. The framework is thus recursive at the data level: construction decisions create feedback fields, feedback improves the reviewer, and reviewer failures can trigger schema or rubric revision.

3.3 SEC Instantiation

Each SEC example contains three test-time fields: `sec_comment`, `company_response`, and `retrieved_snippets` from amended filings. The output is binary. `resolved` means visible amended evidence covers the material request; `unresolved`

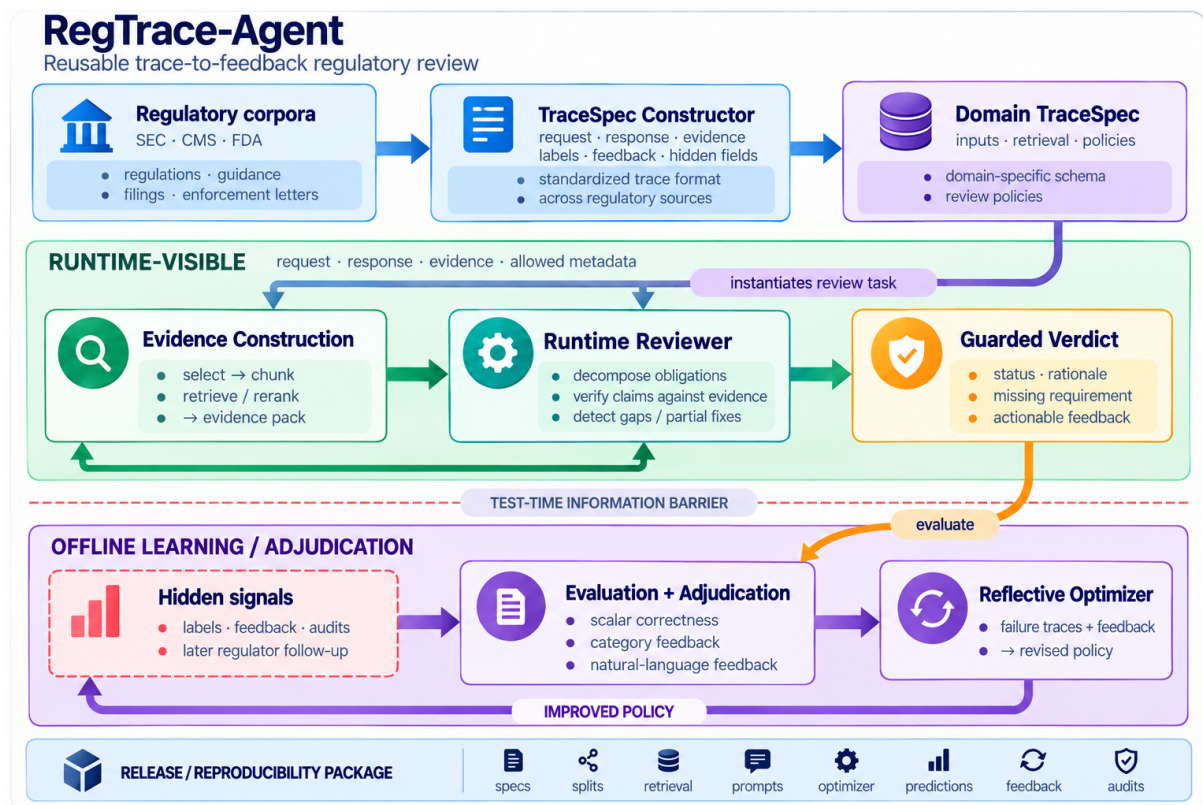


Figure 1: RegTrace-Agent architecture. A TRACESPEC constructor maps each regulatory corpus into request, response, evidence, label, feedback, and hidden-field roles, reshaping the task when required. The runtime reviewer sees only approved inputs and evidence, performs obligation-level review, and emits a guarded verdict. Offline adjudication, audit, and follow-up signals become scalar, category, and written feedback for GEPA-style policy improvement. The release package preserves splits, metadata, retrieval records, prompts, and outputs for reproducibility.

means a named item, quantification, exhibit, document, legal analysis, accounting analysis, or other material requirement remains missing. The label is a visible-evidence judgment, not a legal determination or an inference from absence of later follow-up.

3.4 SEC Trace Construction

Figure 1 shows how trace construction, runtime review, offline feedback learning, and release artifacts fit together. In the SEC instantiation, we harvest UPLOAD and CORRESP materials from EDGAR, group staff comments and company responses into review threads, and select cases where the company claims a disclosure was added, revised, removed, or filed elsewhere. We identify amended filing candidates using filing metadata, target forms mentioned in the review subject, amendment references, page cues, response dates, and response text. The released prototype searches nearby company filings around the response date, excludes UPLOAD/CORRESP, keeps at most eight candidate fil-

ings per row, chunks filing text into 1,800-character windows with 250-character overlap, ranks chunks by salient terms from the SEC comment, response, and review subject, and passes the top three snippets to an LLM reranker/adjudicator. From 589 revision-claim cases, 588 have candidate filings and snippets; 472 remain after requiring visible evidence that directly or partially addresses the SEC request. We then adjudicate whether visible evidence satisfies the request and construct labels plus feedback fields.

The retrieval component gives the agent evidence to inspect. The adjudication and feedback layer gives it something to learn from: what the reviewer asked for, what the company claimed to do, what the visible evidence supports, and what remains missing. These adjudication fields are not shown as ordinary test-time inputs. They are used for optimization feedback, audits, and analysis.

Quantity	Count
Examples	472
SEC review threads	109
Companies	95
Resolved / unresolved	153 / 319
Examples with verified follow-up	236
Issue categories	7
Visible-evidence gap types	8

Table 1: Benchmark scale. Issue categories include other disclosure issues, risk factors, SPAC disclosure, revenue recognition, crypto, non-GAAP, and cybersecurity.

3.5 Scale, Categories, and Splits

RegTrace-SEC contains 472 examples from 109 SEC review threads and 95 companies. The benchmark focuses on difficult revision-claim cases, and the label distribution is operationally realistic: 153 examples are resolved, and 319 are unresolved. Table 1 summarizes the corpus and split regimes.

The corpus covers seven issue categories and eight visible-evidence gap types; Appendix E gives the full distribution. We use grouped, temporal, and topic-holdout regimes. The grouped split holds out entire review threads (229/104/139 train/dev/test). The temporal split trains on 2022–2023 and tests on 2024–2025 (264/64/144). The topic-holdout split tests transfer to held-out issue categories (279/68/125), including crypto and non-GAAP cases absent from training categories.

3.6 From Labels to Feedback

A standard benchmark would stop at binary labels. Our construction preserves the textual structure of each decision. For every example, adjudication records the material SEC request, the company action, evidence relevance, supporting evidence or resolution basis, visible unmet requirement, and a short rationale. This lets us compare increasingly rich optimization signals:

- scalar: correct or incorrect;
- category: correct/incorrect plus coarse issue or gap type;
- full: natural-language feedback describing the SEC request, company action, evidence coverage, and unmet requirement.

Full feedback can state that the evidence discusses combined revenue projections but fails to disclose entity-level operating costs, product pricing, gross

margins, and forecast limitations. This is the reusable decision rule that scalar metrics discard.

3.7 Benchmark Validation and Release Protocol

Because the benchmark label is a visible-evidence judgment, we treat label validity as part of the contribution rather than as a preprocessing detail. We use three checks. First, an independent LLM auditor receives only the test-time fields and predicts whether the visible evidence resolves the SEC request; it does not see the released label, feedback fields, missing-requirement fields, or later SEC follow-up. Second, when a later same-topic SEC comment is available, we classify whether it directly or partially corroborates the visible gap. Third, we provide a stratified human-audit sheet that oversamples independent-audit disagreements, follow-up-corroborated unresolved cases, resolved follow-up controls, and issue-category coverage. Human auditors mark resolved, unresolved, or unclear; the audit is designed to revise or flag benchmark rows rather than to train models.

The release is organized around reproducibility boundaries. Test-time input fields are `sec_comment`, `company_response`, and raw retrieved evidence snippets. Training feedback fields include `feedback_sec_request`, `feedback_company_action`, `feedback_evidence_summary`, and `feedback_unmet_requirement`. Forbidden test-time fields include gold labels, feedback fields, evidence summaries, missing-requirement fields, and later SEC follow-up text. The release includes benchmark JSONL files, frozen grouped/temporal/topic splits, accession identifiers, retrieval metadata, prompts, outputs, cost ledgers, and scripts for regenerating all reported tables.

4 Methods

4.1 Prediction Setup

The prediction model receives the SEC comment, company response, and raw retrieved amended-filing snippets. It returns a binary label and a short rationale. Explicit resolved is scored as resolved, unresolved, partially_resolved, and other insufficient labels are normalized to unresolved. The rationale is required for inspection but not directly rewarded.

4.2 Optimization Ladder

We compare reviewer-training signals rather than evidence access. Baseline prompt is a hand-written reviewer instruction. MIPROv2 is scalar prompt optimization. GEPA-scalar uses reflective prompt optimization with correctness-only feedback. GEPA-category adds coarse category feedback. GEPA-full uses full natural-language adjudication feedback. We also include TF-IDF, a frozen sentence-encoder scoring head, a Jev typed decision gate, and an independent-feedback GEPA ablation for reviewer-facing robustness.

All methods receive the same evidence at test time, isolating the optimization signal rather than evidence access. The key comparison is whether natural-language feedback adds value beyond prompt search itself.

4.3 Implementation Details

All main prompt-optimization runs use openai/gpt-4o-mini as the frozen prediction model with temperature 0 and 500 output tokens. GEPA uses the same model as the reflection model with temperature 0.7 and 1500 reflection tokens, seed 17, max_metric_calls=150, reflection minibatch size 4, Pareto candidate selection, and 4 evaluation threads. MIPROv2 uses the same prediction and prompt models, 8 trials, 4 prompt candidates, and at most 4 labeled and 4 bootstrapped demonstrations. The frozen encoder scorer uses all-MiniLM-L6-v2 with a logistic head over request, response, evidence, and request-evidence matching features. The Jev gate uses a choice question over resolved/unresolved. The evidence reranker/adjudicator also uses gpt-4o-mini at temperature 0. Model aliases rather than dated provider snapshots are recorded in the released artifacts.

4.4 Optimization as Feedback-Conditioned Search

We view each method as search over a prompt or prompt program p . Given an input x and label y , the frozen LLM defines $\hat{y}_p(x) = f_\theta(x; p)$, and optimization seeks a prompt with high validation reward. The methods differ in what they observe after an error. Scalar optimization observes only correctness. Category feedback adds a coarse gap type. Full feedback adds a short critique describing the request, company action, visible evidence, and unmet requirement.

The intuition is information-theoretic. Let C be the latent cause of failure, such as missing quantification or unsupported revision claim. Full feedback should help when it lowers uncertainty about C , because the optimizer can propose a more targeted semantic edit. This is not a general guarantee: a noisy or misaligned language channel can hurt. The hypothesis of RegTrace-Agent is that review traces are unusually well suited to language feedback because the failure causes are naturally written as reviewer critiques.

4.5 What the Optimizer Sees

During optimization, each candidate prompt is evaluated on development examples. When a prediction is incorrect, the metric returns feedback in one of three forms:

```
scalar: incorrect
category: incorrect;
gap_type=missing_specific_disclosure
full: incorrect. The SEC requested
X. The company claimed Y. The visible
evidence shows Z, but does not show
required item W.
```

Full feedback summarizes the adjudicated relationship between the SEC request, response, and retrieved evidence for optimization examples only. It does not reveal hidden evidence at test time; any advantage must come from a better learned checklist for visible-evidence review.

5 Experiments

The experiments are organized around five claims. First, amended evidence is necessary for the task to be meaningful. Second, full natural-language feedback is the strongest reviewer-training signal under fixed test-time context. Third, the learned policy is not a narrow random-split artifact. Fourth, the framework transfers by reconstructing domain-specific traces rather than copying an SEC prompt. Fifth, later SEC follow-up provides external evidence that visible-evidence labels track real regulator behavior. Figures 2–6 visualize these claims; the tables report the exact metrics used for reproducibility.

5.1 Evidence Ablation

We first test whether amended-filing evidence matters. Table 2 evaluates a response-only condition, a raw-evidence condition, and an oracle evidence summary condition on the grouped held-out test set.

Input	Acc.	Ma-F1	Res-F1	Unres-F1
Response only	0.712	0.575	0.333	0.817
+ raw snippets	0.784	0.748	0.651	0.844
+ oracle summary	0.964	0.956	0.938	0.975

Table 2: Evidence ablation. Adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle summary is an upper bound, not a deployable baseline.

Response-only prompting cannot verify whether a revision claim is supported. Raw snippets nearly double resolved F1, while the oracle summary shows that much of the remaining difficulty is interpreting snippets against a specific request. Figure 2 is the first core result: the benchmark is not merely a response-quality task, but an evidence-sufficiency task.

5.2 Main Context-to-Feedback Results

Table 3 reports the main reviewer-training ladder on the frozen grouped split. All methods receive the same test-time context.

The majority baseline has high accuracy because unresolved examples dominate, but zero resolved F1. The handwritten prompt is conservative: it has perfect unresolved precision but misses most gaps. MIPROv2 improves substantially, GEPA-scalar improves recall, and GEPA-category is too coarse in this run. GEPA-full is strongest overall, improving macro-F1 by 10.4 points over MIPROv2 and 8.1 points over GEPA-scalar. Figure 3 makes the central comparison explicit: with evidence held fixed, the difference is the richness of the optimization signal. Full feedback trains the reviewer to compare requested obligations against visible evidence instead of merely becoming more or less skeptical.

Table 4 addresses two alternate explanations and one deployment question. A TF-IDF logistic regression classifier trained on the same raw evidence reaches only 0.523 macro-F1, so lexical supervision does not explain the GEPA-full gain. A frozen sentence encoder with a trained logistic scoring head reaches 0.667 macro-F1 without generation, and a Jev typed decision gate reaches 0.722, showing that RegTrace-Agent can support cheap triage scorers, but still trails the full feedback-trained reviewer. On the same grouped held-out test set ($N = 139$), the guarded reviewer improves over a monolithic prompt under both gpt-4o-mini and gpt-5.4-mini, suggesting that obligation-evidence structure matters across backbones. To test possible feedback leakage, we re-

generate full textual feedback with an independent gpt-5.4-mini writer that sees only test-time fields plus the gold label, not the original adjudication rationales; GEPA with this feedback reaches 0.750 macro-F1, nearly matching the original 0.756.

We also compute five-fold out-of-fold results over review-thread groups. Mean macro-F1 is 0.698 for GEPA-full, 0.647 for MIPROv2, 0.609 for GEPA-scalar, 0.545 for GEPA-category, and 0.446 for the handwritten baseline. The fold standard deviations are nontrivial (0.151 for GEPA-full and 0.121 for MIPROv2), so we treat the single grouped split as a headline comparison and the out-of-fold analysis as stability evidence rather than as a deterministic margin claim.

5.3 Why Full Feedback Helps

The advantage of GEPA-full is not simply risk aversion. Jev reaches very high unresolved recall (0.958) but lower resolved F1 (0.576), making it useful as a triage gate that flags likely unresolved responses. GEPA-full has lower unresolved recall but higher macro-F1, because it better balances skepticism with evidence sufficiency. It learns when a company revision claim is unsupported, but also when visible amended evidence already satisfies the SEC request.

The mechanism is visible in the feedback channel. Scalar feedback says only that a decision was wrong; category feedback names a coarse gap. Full feedback states what the SEC requested, what the company claimed, what the amended evidence showed, and what requirement remained unmet. This gives the optimizer a reusable review trace rather than a bare reward. The resulting prompt evolution shifts from generic caution to obligation-level comparison: identify material SEC request elements, inspect amended evidence for named disclosures, quantification, exhibits, and legal or accounting analysis, and distinguish partial disclosure from resolution.

The trade-off is cost and calibration. GEPA-full requires adjudicated natural-language feedback and offline optimization, so it is not the cheapest runtime gate. It can also become over-skeptical when evidence substantially satisfies a request but does not present the answer in the exact form the prompt expects. We therefore view Jev and frozen encoders as high-throughput first-pass gates, and GEPA-full as the deeper reviewer for difficult or high-impact cases.

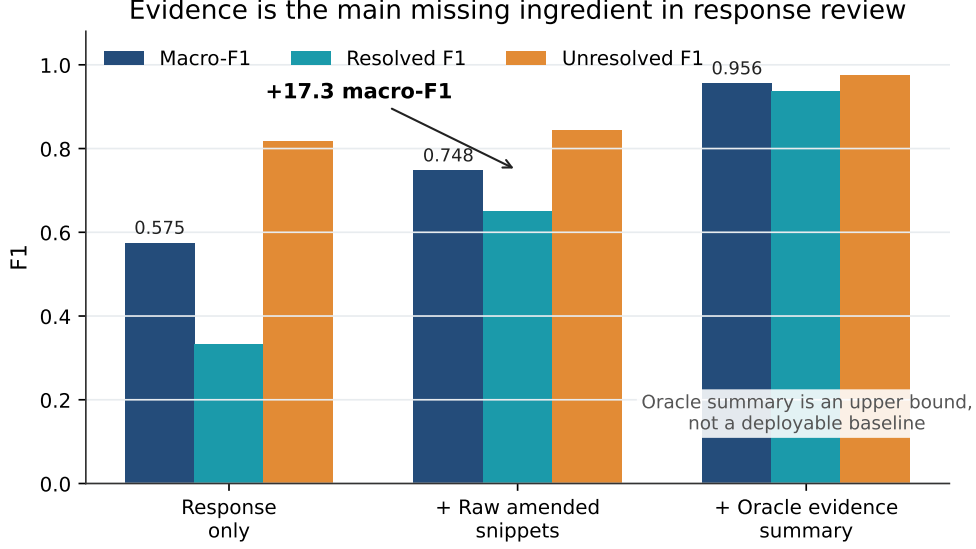


Figure 2: Evidence ablation. Response-only review cannot verify revision claims; adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle evidence summary shows that better evidence selection and interpretation remain the main bottleneck.

Method	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Majority unresolved	0.691	0.409	0.000	0.817	0.691	1.000
Baseline prompt	0.424	0.402	0.518	0.286	1.000	0.167
MIPROv2	0.655	0.652	0.619	0.684	0.929	0.542
GEPA-scalar	0.705	0.675	0.577	0.773	0.824	0.729
GEPA-category	0.597	0.595	0.562	0.627	0.870	0.490
GEPA-full	0.784	0.756	0.674	0.839	0.867	0.812

Table 3: Frozen grouped-split leaderboard. GEPA-full gives the best macro-F1 and unresolved F1 under the same raw-evidence test-time context.

5.4 Reviewer-Structure Ablation

The optimization ladder tests feedback richness. We also ask whether reviewer structure matters. Table 5 compares a monolithic judgment, an obligation-guided judgment, and a guarded verifier that requires every material obligation to be visibly satisfied, with GEPA-full as the optimized-prompt reference.

Simply listing obligations is not enough; the gain comes from the guard that requires visible support for each material request. We report this as a policy-distillation prototype rather than the primary optimizer comparison, since it adds human-designed structure after prompt-evolution analysis.

5.5 Feedback Component Ablation

Table 6 holds test-time input fixed and changes only optimization-time feedback.

Request-action-only feedback induces aggressive skepticism and the highest unresolved recall, but sacrifices resolved F1. Full feedback teaches

both sides of the boundary: unmet requirements for unresolved cases and supporting evidence for resolved cases.

5.6 Temporal and Topic Generalization

Table 7 shows that the optimized prompt transfers beyond the grouped split. In the temporal setting, GEPA-full trains/tunes on 2022–2023 and tests on 2024–2025 material. In the topic setting, test examples include crypto and non-GAAP cases absent from training issue categories.

The topic-holdout result argues against a curated topic explanation. As summarized in Figure 4, the full-feedback policy transfers both forward in time and into held-out issue categories. Per-topic breakdowns show GEPA-full improves over MIPROv2 on crypto, non-GAAP, revenue recognition, risk factors, and broad other-disclosure cases, with largest gains in revenue recognition and non-GAAP.

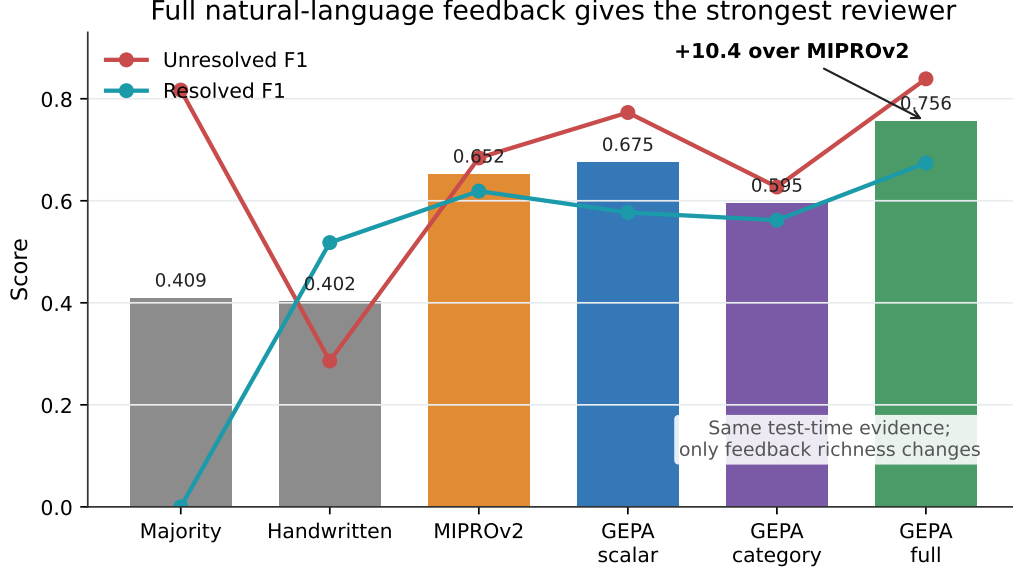


Figure 3: Feedback ladder on the grouped SEC test set. All prompt-optimization methods receive the same test-time evidence. GEPA-full gives the strongest macro-F1 and the best balance between resolved and unresolved cases, suggesting that written feedback teaches an obligation-evidence review policy that scalar reward does not expose.

Check	Condition	Macro-F1
Lexical baseline	TF-IDF, response only	0.523
Lexical baseline	TF-IDF, raw snippets	0.523
Scoring head	Frozen encoder, raw snippets	0.667
Decision gate	Jev typed choice, raw snippets	0.722
Reviewer structure	4o-mini monolithic / guarded	0.638 / 0.808
Reviewer structure	5.4-mini monolithic / guarded	0.612 / 0.732
Feedback source	GEPA-full, original feedback	0.756
Feedback source	GEPA-full, independent feedback	0.750
Diagnostic upper bound	TF-IDF, oracle evidence summary	0.858

Table 4: Robustness checks on the grouped held-out test set unless noted. The raw-snippet TF-IDF baseline shows that lexical supervision does not explain the main result. Frozen encoder and Jev scorers are cheap triage components. Multi-LLM reviewer runs show that guarded obligation-evidence structure improves over monolithic prompting under two backbones. The independent-feedback ablation shows that GEPA-full does not depend on reusing the original adjudication rationales.

Reviewer	Macro-F1	Res-F1	Unres-F1
Monolithic reviewer	0.645	0.614	0.676
Obligation-guided reviewer	0.619	0.613	0.624
GEPA-full optimized prompt	0.756	0.674	0.839
RegTrace guarded verifier	0.801	0.727	0.874

Table 5: Reviewer-structure ablation. The guarded verifier is a policy-distillation prototype derived from the obligation-evidence policy exposed by GEPA-full prompt evolution.

6 Cross-Regulatory Adaptation: CMS-2567

We next ask whether RegTrace-Agent is only a curated SEC benchmark or whether the same context-to-feedback-review pattern can be instantiated in a different regulatory domain. We use CMS-2567 nursing-home Statements of Deficiencies and Plans of Correction. The object of review is different from SEC filings: a facility receives a deficiency narrative, submits a Plan of Correction (POC), and the reviewer asks whether the plan materially addresses resident-specific correction, affected

population, systemic prevention, monitoring, timing, and accountability. The test-time input is deficiency_text, plan_of_correction_text, ftag, and scope_severity. Feedback-only fields record covered elements, missing or weak elements, and a reviewer rationale; correction-outcome meta-data is hidden and not used as a gold label because most cases eventually become corrected.

This experiment is intentionally framed as adaptation, not direct zero-shot transfer. The SEC optimized prompt checks amended disclosure against named SEC requests. CMS POC review requires a different calibration: a plan can be adequate without proving that every future monitoring step has already succeeded, but it is not adequate if it merely lists education and audits while omitting a cited resident-specific or systemic deficiency.

We construct a 500-example CMS pilot benchmark with grouped train/dev/test splits by source document (270/90/140 examples). Labels map

Feedback	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Full feedback	0.784	0.756	0.674	0.839	0.867	0.812
No evidence rationale	0.712	0.682	0.583	0.780	0.826	0.740
No unmet requirement	0.741	0.708	0.609	0.806	0.833	0.781
Request + action only	0.784	0.724	0.595	0.853	0.806	0.906

Table 6: Feedback component ablation. Request-action feedback induces aggressive unresolved detection, while full feedback gives the best macro-F1 by calibrating skepticism with evidence rationales.

Split	Method	Acc.	Macro-F1	Unres-F1	Unres-R
Time	Baseline	0.451	0.434	0.336	0.206
	MIPROv2	0.521	0.519	0.489	0.340
	GEPA-scalar	0.653	0.650	0.679	0.546
	GEPA-full	0.826	0.791	0.877	0.918
Topic	Baseline	0.440	0.440	0.426	0.274
	MIPROv2	0.576	0.565	0.635	0.484
	GEPA-scalar	0.528	0.524	0.569	0.411
	GEPA-full	0.784	0.734	0.849	0.800

Table 7: Temporal and topic-holdout generalization. GEPA-full transfers substantially better than scalar optimization.

adequate to adequate and both partial and inadequate to not_adequate. The grouped test set contains 37 adequate and 103 not-adequate examples. These labels are LLM-adjudicated and therefore weaker than the SEC benchmark; we use the study to test framework portability rather than to claim a fully validated second benchmark.

Table 8 shows the central pattern. A direct generic CMS prompt is already strong (0.694 macro-F1), which means POC adequacy is closer to ordinary regulatory reading than SEC amended-evidence verification. The SEC-style structured reviewer, however, is too conservative: it predicts only two adequate cases and reaches 0.449 macro-F1 despite high nominal accuracy from class imbalance. This is a useful negative result. RegTrace Agent does not mean transferring a frozen SEC checklist. It means re-instantiating the same loop with domain-appropriate feedback.

Under that adaptation, GEPA-full improves the minimal CMS seed from 0.642 to 0.723 macro-F1, beats scalar feedback by 5.9 points, and beats the direct generic prompt by 2.9 points. The gain is not pure skepticism. GEPA-full predicts fewer adequate cases than the generic prompt, but it also preserves high adequate recall while improving not-adequate F1. The full feedback teaches the reviewer when standard POC language is enough and when it is boilerplate that does not address the cited deficiency.

What transfers. The reusable part is not a prompt string. It is the context-to-feedback mech-

anism: identify the reviewer request or deficiency, map the response to material obligations, use adjudicated text to expose covered and missing elements, and optimize a reviewer around that feedback. SEC review learns “do not accept a revision claim unless amended evidence contains the named disclosure.” CMS review learns “substantial adequacy is enough when the plan covers immediate correction, affected population, systemic prevention, monitoring, and timing, but generic education/audit language is not enough when a cited deficiency component is omitted.”

7 Third-Domain Stress Test: FDA Warning Letters

The CMS study tests positive adaptation. We also test whether the construction layer can avoid a bad adaptation. FDA warning-letter data provides public warning letters and closeout letters, but public company response letters are almost absent in our snapshot. A fixed SEC-style framework would force this into a request-response-evidence task and produce a weak benchmark. RegTrace Agent instead emits a TRACESPEC decision of reshape-before-benchmark: the observable task is warning-to-closeout trace verification, not response adequacy.

We construct a minimal FDA contrastive benchmark from 400 linked warning-closeout pairs. Each warning receives one positive candidate, its true linked closeout letter, and one hard negative closeout letter, preferably sampled from the same

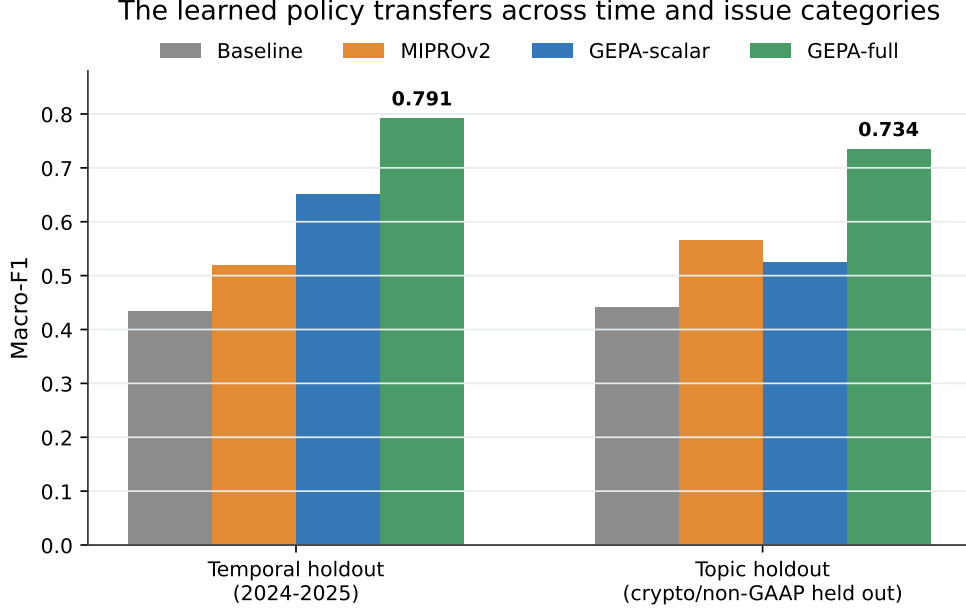


Figure 4: Temporal and topic-holdout generalization. GEPA-full remains strongest when trained on earlier years and tested on 2024–2025, and when crypto/non-GAAP topics are held out from training categories.

Method	Seed	Feedback	Acc.	Macro-F1	Adeq-F1	Not-Adeq-F1
Direct generic prompt	generic	none	0.707	0.694	0.631	0.757
Minimal seed	minimal	none	0.650	0.642	0.588	0.696
GEPA-scalar	minimal	correctness only	0.671	0.663	0.610	0.716
GEPA-category	minimal	coarse category	0.643	0.635	0.583	0.688
GEPA-full	minimal	written POC feedback	0.743	0.723	0.647	0.798
Structured SEC seed	SEC-style	none	0.736	0.449	0.051	0.847
Structured GEPA-full	SEC-style	written POC feedback	0.771	0.584	0.304	0.863

Table 8: CMS-2567 adaptation on the grouped 140-example test set. The SEC-style checklist does not transfer verbatim: it predicts almost everything as not adequate. Starting from a lightweight CMS reviewer, full written feedback beats scalar/category feedback and a direct generic prompt.

Reviewer	Acc.	Ma-F1	Mat. F1	Wrong F1
Generic	0.919	0.918	0.912	0.925
RegTrace strict	0.881	0.880	0.865	0.894
RegTrace calibrated	0.950	0.950	0.947	0.952

Table 9: FDA warning-to-closeout trace verification on a 160-example held-out test set using gpt-4o-mini. The strict RegTrace reviewer catches all wrong closeouts but over-rejects true closeouts. A domain-calibrated reviewer preserves mismatch detection while recovering true matches.

730 product and subject. This yields 800 examples
731 with grouped train/dev/test splits by warning URL
732 (480/160/160). The held-out test set is balanced,
733 and 357 of 400 negatives in the full benchmark are
734 same-product and same-subject wrong closeouts.
735 Table 9 shows the recursive behavior we want
736 from the framework. The first strict reviewer is
737 too skeptical: it catches every mismatched close-

out but rejects 19 true matches, often because FDA 738
closeout letters are short and do not repeat every 739
violation. The calibrated reviewer changes the 740
rubric, not the hidden data. It treats firm iden-741
tity, CMS/reference number, warning date, prod-742
uct or violation family, issuing office, and explicit 743
corrective-action evaluation as the trace criteria,744
while not requiring the closeout to restate the full 745
warning letter. This improves macro-F1 from 0.880 746
to 0.950 and also exceeds the generic prompt. FDA 747
therefore serves as a compact third-domain demon-748
stration of adaptive construction: when a corpus 749
lacks a response channel, RegTrace-Agent reshapes 750
the task and then calibrates the reviewer to the new 751
document genre. 752

RegTrace transfers as a construction-and-feedback pattern, not a copied SEC prompt

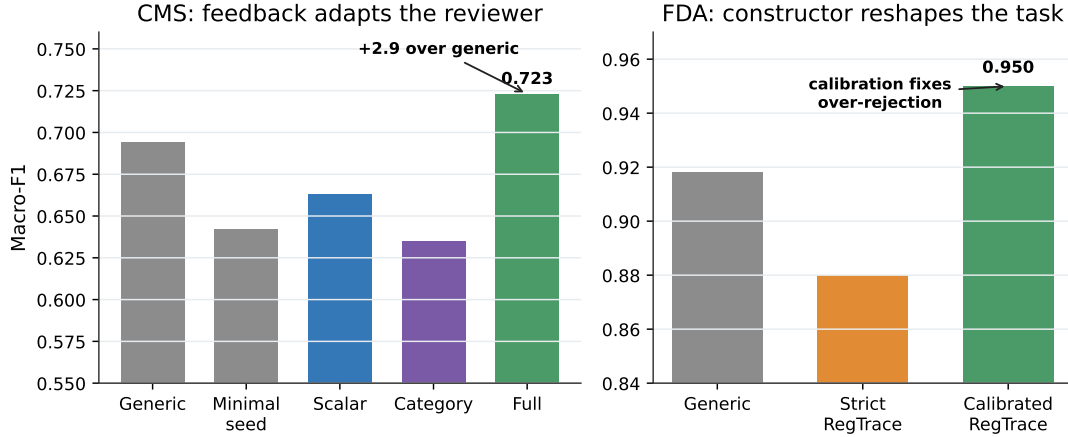


Figure 5: Cross-regulatory adaptation. CMS shows that full written feedback can adapt the reviewer to a different response object, Plans of Correction. FDA shows that the construction layer can reshape the task when a corpus lacks public company response letters. Together, these results support the claim that RegTrace-Agent transfers as a construction-and-feedback pattern, not as a copied SEC prompt.

Group	N	D/P	Rate
Unres. + f/u	160	109	68.1%
Res. + f/u	76	7	9.2%

Table 10: Real follow-up corroboration. Later same-topic SEC comments directly or partially corroborate unresolved visible-evidence labels far more often than resolved labels.

8 External SEC Follow-Up Corroboration

Visible-evidence labels come from an adjudication pipeline, so we test whether they align with later regulator behavior. Verified same-topic next-round SEC follow-up is not the primary label, but it is a useful corroboration signal because it can restate or narrow an unresolved gap.

We additionally audit all 472 examples with an independent gpt-5.4-mini auditor that receives only the SEC comment, company response, and raw retrieved evidence snippets, not the benchmark label, feedback fields, evidence summaries, missing-requirement notes, or later SEC follow-up. The audit agrees with the benchmark label in 345 of 472 cases, reaching 0.731 agreement accuracy and 0.715 macro-F1. We report this as an audit of visible-evidence label consistency, not as expert legal/accounting validation.

Among 236 examples with verified follow-up, unresolved cases are corroborated at 68.1%, while resolved cases are corroborated at only 9.2% (Table 10). Follow-up availability itself is balanced

across labels (49.7% for resolved and 50.2% for unresolved; Fisher exact $p = 1.0$), so this is not merely an availability artifact. Figure 6 presents this as a label-validity result rather than as an extra model score.

Following reviewer-style paired analysis, we also treat later follow-up as a noisy external reference on a clean 220-example subset. The benchmark label reaches 0.756 macro-F1 against this signal, compared with 0.590 for five-fold out-of-fold GEPA-full predictions and 0.460 for the handwritten baseline. We therefore use follow-up primarily as label-validity evidence, not as proof that every GEPA-full win is separately corroborated beyond the unresolved base rate.

In one expense-disclosure case, the SEC asked for material components and trend discussion. The company claimed a revision, but retrieved evidence showed only a line item and totals; GEPA-full predicted unresolved, and the next-round SEC follow-up again requested the component breakout. SPAC transaction-background cases show the same pattern for missing negotiation details.

9 Prompt Evolution and Case Studies

9.1 From Terse Prompt to Reviewer Policy

Prompt evolution is the most visible evidence that RegTrace-Agent is not simply tuning a label instruction. In both domains, the seed prompt is deliberately weak. SEC starts from: “Classify whether a

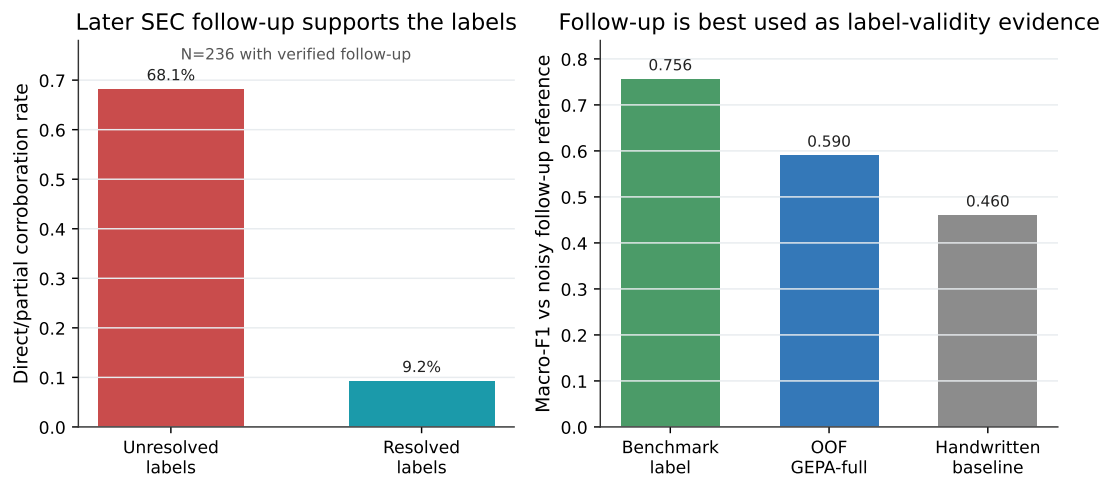


Figure 6: External label-validation evidence from later SEC follow-up. Unresolved visible-evidence labels are much more often directly or partially corroborated by later same-topic SEC comments than resolved labels. A paired noisy-reference analysis also shows that the benchmark label aligns with follow-up better than model predictions, supporting use of follow-up as label-validity evidence rather than as the primary gold label.

company resolved an SEC comment using the visible evidence.” CMS starts from: “Classify whether a CMS Plan of Correction adequately resolves the deficiency.” Full feedback then rewrites these seeds into different reviewer policies because the feedback describes different kinds of gaps.

The component analysis in Table 12 shows the same pattern quantitatively. Scalar and category feedback learn generic caution: company claims need support. Full feedback more consistently induces named-request matching and missing-item reasoning, shifting the policy from generic skepticism to obligation-level comparison: request → response action → visible evidence or correction plan → remaining gap.

9.2 What the Feedback Looks Like

The feedback channel is not a hidden test-time input. It is an optimization-time critique. A SEC unresolved example can return: the SEC requested individual-company forecasts and material assumptions; the company claimed to revise the forecast section; the amended evidence shows combined revenue projections; operating costs, pricing, margins, and forecast limitations remain missing. A CMS not-adequate example can return: the plan covers refrigerator temperature logs but does not address the same deficiency narrative’s fall-risk care-plan failure or RN coverage/PBJ issue. These critiques are exactly the information a scalar reward loses.

9.3 SEC Case: Combined Forecasts Are Not Entity-Level Assumptions

One SEC comment asked whether the board received VSee and iDoc projections on an individual-company basis and requested entity-level assumptions, including growth rates, operating costs, product pricing, gross margins, and forecast limitations. The company responded that it revised the prospective-financial-information section. The retrieved amended evidence gave combined revenues for 2022 and 2023 and some context about who prepared the projections. The baseline, GEPA-scalar, and GEPA-category predicted resolved; GEPA-full predicted unresolved.

The GEPA-full rationale is qualitatively different. It does not stop at seeing a revised forecast paragraph. It maps the SEC request to named missing requirements: individual-company basis, material assumptions, operating costs, pricing, gross margins, and limitations. This is the regulator-style reasoning we want the system to learn: a response can be responsive in topic but unresolved in obligation coverage.

9.4 SEC Case: Stablecoin Definition Without the Requested Contrast

In a crypto disclosure case, the SEC asked the company to clarify what it meant by “stable cryptocurrency” as opposed to an “unstable” cryptocurrency. The company said it changed the term to “stablecoin” and described stablecoins as fiat-pegged assets. The amended evidence defined stablecoins

Policy	What the prompt learns	Representative optimized language
SEC seed	Generic evidence check.	“Classify whether a company resolved an SEC comment using the visible evidence.”
SEC GEPA-full	Compare named SEC requests against amended evidence; distrust unsupported revision claims; find missing disclosures, quantification, exhibits, and legal/accounting analysis.	“Carefully compare the SEC request against the company’s response and the amended evidence” and look for “specific items,” “quantification,” “detailed analysis,” and “visible unmet requirements.”
CMS seed	Generic POC adequacy check.	“Classify whether a CMS Plan of Correction adequately resolves the deficiency.”
CMS GEPA-full	Judge substantial POC adequacy: resident-specific correction, affected population, systemic prevention, staff education, monitoring, timing, accountability, and material omissions tied to the deficiency narrative.	Assess whether the POC provides “clear, actionable steps,” “resident-specific corrections, systemic changes, staff education, and ongoing monitoring,” and avoid relying on “vague or boilerplate language.”

Table 11: Prompt evolution under full feedback. The SEC and CMS optimized prompts share the same context-to-feedback mechanism, but they learn different domain policies. SEC focuses on visible amended disclosure; CMS focuses on whether a remediation plan materially covers the deficiency.

Method	Claim insuff.	Evidence required	Request match	Missing items
GEPA-full	2/2	2/2	2/2	2/2
GEPA-scalar	2/2	2/2	0/2	0/2
GEPA-category	2/2	2/2	0/2	0/2
Req.+action only	2/2	2/2	2/2	2/2

Table 12: Rule presence in SEC optimized prompt candidates. Full feedback more consistently induces named-request matching and missing-item reasoning.

865 and mentioned examples such as USDC and USDT.
866 Baseline, MIPROv2, scalar GEPA, and category
867 GEPA predicted resolved; GEPA-full predicted
868 unresolved.

869 The learned policy catches a subtle gap: the evi-
870 dence defines the stable side but does not directly
871 provide the requested contrast with unstable cryp-
872 tocurrency. This is not a lexical trick; it is request
873 matching. The optimized SEC prompt tells the re-
874 viewer to identify “specific items or information
875 requested by the SEC” and verify that amended
876 evidence covers “all requested aspects.”

877 9.5 CMS Case: Full Feedback Is Not Just 878 More Skeptical

879 CMS also shows why full feedback is not merely
880 an unresolved detector. In one F0689 resident-
881 safety case, the deficiency was an unsecured treat-
882 ment cart. The POC gave an immediate teach-
883 able moment, identified all residents as poten-
884 tially affected, trained nursing staff on locked
885 cart storage, reviewed cart storage capacity, and
886 created weekly/monthly accident-hazard audits
887 with QAPI review. The generic prompt predicted

not_adequate, arguing that the plan lacked a di-888
rect assurance that carts would always be secured.889
GEPA-full predicted adequate. 890

The full feedback for similar adequate cases 891
teaches calibration: substantial adequacy is enough 892
when the plan covers immediate correction, af-893
fected population, systemic prevention, monitoring, 894
and timing. This is the opposite of the SEC-style 895
over-skeptical checklist. It shows that full feed-896
back can also reduce false positives by teaching the 897
reviewer not to demand more than the regulatory 898
object requires. 899

900 9.6 CMS Case: Boilerplate Audits Do Not 901 Cover a Multi-Part Deficiency

In another F0689 case, the deficiency narrative 902
combined several issues: fall-risk care planning 903
for a resident, RN eight-hour coverage/PBJ report-904
ing, and refrigerator temperature monitoring. The 905
POC addressed refrigerator gauges, logs, policy 906
education, and audits. GEPA-scalar and GEPA-907
category predicted adequate. GEPA-full predicted 908
not_adequate, because the POC did not visibly 909
correct the fall-risk care-plan problem or the RN 910
coverage/PBJ problem. 911

This case is the CMS analogue of the SEC pro-912
jection example. The response contains plausible 913
compliance language and a real corrective action,914
but only for one component of the deficiency. Full 915
feedback teaches the model to decompose the de-916
ficiency narrative and ask whether each material 917
component has a corresponding corrective action. 918

919 9.7 Known Failure Mode

920 The same mechanism can over-require. In a SEC
921 ownership-percentage case, retrieved evidence con-
922 tained redemption-scenario calculations and the
923 benchmark label was resolved. GEPA-full pre-
924 dicted unresolved because it wanted a more ex-
925 plicit cover-page presentation. In CMS, the analo-
926 gous risk is treating a practical POC as incomplete
927 because it lacks idealized audit escalation language.
928 This is why the guarded verifier and CMS adapta-
929 tion results should be framed as review-assistant
930 policies with calibrated feedback, not as legal or
931 clinical ground truth.

932 10 Discussion

933 The main result is not simply that one optimizer
934 beats another. RegTrace-Agent turns a recur-
935 ring compliance pattern into an offline feedback-
936 learning system: collect the trace, retrieve the re-
937 vised artifact, check obligations against visible ev-
938 idence, issue a guarded verdict, and recycle adju-
939 dicated failures as feedback. The learned reviewer
940 can be used directly, distilled into a guarded veri-
941 fier, or converted into lightweight scoring heads for
942 high-throughput triage. The output is a traceable
943 unresolved-gap rationale, not only a binary label.

944 Language feedback helps because it preserves
945 decision structure. A scalar signal says only that
946 a verdict failed; full feedback says which named
947 request, company action, amended snippet, and
948 remaining obligation were mismatched. The opti-
949 mized prompt therefore becomes an interpretable
950 policy: extract material requests, distrust unsup-
951 ported revision claims, inspect evidence for named
952 items, avoid over-requiring, and distinguish partial
953 disclosure from resolution.

954 SEC correspondence is the public setting where
955 we can observe request, response, revised filing,
956 and later reviewer behavior, but the trace structure
957 is broader. Audit remediation, regulatory report-
958 ing, lending-document stipulations, data-quality
959 exceptions, and internal compliance monitoring
960 also contain request-response-evidence loops. The
961 transferable claim is that such traces can support
962 retrieval, obligation checking, adjudication feed-
963 back, optimized reviewing, and hard-case mining;
964 the limitation is that high-recall language-feedback
965 policies can become over-skeptical.

11 Conclusion

966

We introduced RegTrace-Agent, a context-to-967
feedback framework for evidence-grounded re-968
sponse review, and instantiated it as RegTrace-SEC 969
using public SEC traces. Across grouped, tem-970
poral, and topic-holdout evaluations, full natural-971
language feedback yields the strongest reviewer.972
Evidence ablations, prompt evolution, and SEC 973
follow-up corroboration suggest that financial NLP 974
benchmarks can move from static labels toward 975
feedback-rich review systems that preserve the rea-976
sons behind expert decisions. 977

Limitations and Data Statement

978

This study is conditioned on visible retrieved ev-979
idence, so missed snippets can affect labels. The 980
SEC benchmark is LLM-adjudicated, with inde-981
pendent audit and external follow-up corroboration 982
but without a full expert legal/accounting audit. A 983
full independent gpt-5.4-mini audit over all 472 984
examples, using only test-time fields, reaches 0.731 985
agreement accuracy and 0.715 macro-F1 against 986
the released labels. Same-topic SEC follow-up is 987
a noisy behavioral signal, not ground truth. We 988
frame labels as visible-evidence judgments rather 989
than legal determinations. The CMS-2567 study 990
is weaker and should be read as a cross-regulatory 991
adaptation pilot: its 500 labels are primary LLM 992
adjudications, and the partial boundary is noisier 993
than the SEC visible-evidence label. 994

An anonymized release will be provided 995
at [https://anonymous.4open.science/r/](https://anonymous.4open.science/r/regtrace-sec-XXXX/) 996
[regtrace-sec-XXXX/](https://anonymous.4open.science/r/regtrace-sec-XXXX/). The release includes 997
benchmark JSONL files, frozen split files, acces-998
sion identifiers, retrieval metadata, snippets, labels, 999
feedback fields, prompt states, model outputs, cost1000
ledgers, and scripts for generating the main tables1001
We also include a stratified human-audit sheet1002
and guideline so later users can inspect or revise1003
high-uncertainty labels without mixing audit-only1004
fields into model inputs. 1005

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1138	benchmark on a hybrid of tabular and textual content		
1139	in finance. In <i>Proceedings of the 59th Annual Meet-</i>		
1140	<i>ing of the Association for Computational Linguistics</i> .		
1141	Association for Computational Linguistics.		
1142	A Full Optimized Prompts		
1143	The main paper quotes short excerpts from opti-		
1144	mized prompts. Below are the full GEPA-full op-		
1145	timized candidates exposed by the final grouped-		
1146	split run. The seed prompt was: “Classify whether		
1147	a company resolved an SEC comment using the		
1148	visible evidence.”		
1149	A.1 GEPA-full Optimized Candidate 1		
1150	You are tasked with classifying whether a com-		
1151	pany has resolved a comment from the SEC based		
1152	on the provided evidence. Follow these instruc-		
1153	tions carefully:		
1154	1. Input Format: You will receive three types of		
1155	inputs: <code>sec_comment</code> , <code>company_response</code> , and		
1156	<code>amended_evidence</code> .		
1157	2. Classification Criteria: Determine if the com-		
1158	pany has fully resolved the SEC comment by ana-		
1159	lyzing the company response in conjunction with		
1160	the amended evidence. If the response satisfies		
1161	the SEC’s request and the evidence supports this		
1162	claim, classify the comment as resolved. If the re-		
1163	sponse does not meet the SEC’s request or if there		
1164	are still outstanding issues, classify the comment		
1165	as unresolved.		
1166	3. Evaluation Process: Carefully compare the		
1167	SEC request against the company’s response and		
1168	the amended evidence. Look for specific items or		
1169	information requested by the SEC, quantification		
1170	or detailed analysis that meets SEC requirements,		
1171	references to regulatory guidance or previous de-		
1172	cisions, and clear explanations of how the com-		
1173	pany adjusted disclosures or practices. Take note		
1174	of visible unmet requirements, partial fixes, or		
1175	lack of supporting documentation in the amended		
1176	evidence.		
	4. Output Format: Include a reasoning section,	1177	
	a label indicating resolved or unresolved, and a	1178	
	concise reason.	1179	
	5. Feedback Incorporation: Be vigilant about	1180	
	claims made by the company; ensure that	1181	
	amended evidence corroborates these claims.	1182	
	Identify instances where the response may par-	1183	
	tially address SEC concerns but still falls short of	1184	
	complete compliance.	1185	
	A.2 GEPA-full Optimized Candidate 2	1186	
	You are tasked with classifying whether a com-	1187	
	pany has resolved a comment issued by the SEC	1188	
	based on provided evidence. Follow these specific	1189	
	instructions for the task:	1190	
	1. Input Format: You will receive <code>sec_comment</code> ,	1191	
	<code>company_response</code> , and <code>amended_evidence</code> .	1192	
	The SEC comment is the original request for clar-	1193	
	ification or additional information. The company	1194	
	response indicates what changes were made or	1195	
	explanations provided. The amended evidence	1196	
	contains snippets from the amended filing or other	1197	
	documents that support the response.	1198	
	2. Classification Criteria: Determine whether the	1199	
	company’s response and amended evidence ade-	1200	
	quately address the SEC’s concerns as outlined	1201	
	in the original comment. A response is resolved	1202	
	if it fully satisfies the requirements of the SEC	1203	
	comment, including clear explanations, specific	1204	
	details, and quantifications when necessary. A	1205	
	response is unresolved if there are still gaps or	1206	
	missing elements in the company’s explanation or	1207	
	if amended evidence does not address all parts of	1208	
	the SEC comment.	1209	
	3. Evaluation Process: Carefully analyze the SEC	1210	
	comment and identify the specific details or expla-	1211	
	nations requested. Review the company response	1212	
	to see if the company acknowledged the comment	1213	
	and provided a meaningful response. Examine	1214	
	amended-evidence snippets to confirm that they	1215	
	support the company’s claims and cover all re-	1216	
	quested aspects of the SEC comment. Look for	1217	
	direct references in the amended evidence that	1218	
	address the SEC’s specific requests, such as own-	1219	
	ership structures, accounting adjustments, or en-	1220	
	forcement risks.	1221	
	4. Labeling: If the company has fully addressed	1222	
	the SEC’s concerns with adequate evidence, label	1223	
	the response resolved. If there are omissions or	1224	
	inadequacies in addressing SEC requests, label it	1225	
	unresolved.	1226	
	5. Justification: Provide clear and concise rea-	1227	
	soning that cites specific elements from the SEC	1228	
	comment, company response, and amended evi-	1229	
	dence. Highlight specific unmet requirements or	1230	
	gaps in the evidence that contribute to an unre-	1231	
	solved classification.	1232	
	6. Feedback Incorporation: Pay attention to feed-	1233	
	back from previous examples to identify common	1234	
	pitfalls, such as failing to compare the SEC re-	1235	
	quest against the evidence adequately or missing	1236	
	key elements in the response.	1237	
	B CMS Optimized Prompt	1238	
	The CMS seed prompt was: “Classify whether a	1239	
	CMS Plan of Correction adequately resolves the	1240	

deficiency.” The full-feedback CMS run under the 270/90/140 grouped split produced a learned reviewer behavior whose outputs emphasize resident-specific correction, systemic prevention, education, monitoring, timing, and material omissions. We report the final reviewer instruction used for the CMS full-feedback policy below. Unlike the SEC appendix above, this is a paper-facing distilled instruction for the optimized CMS reviewer rather than a raw two-candidate state dump.

Classify the adequacy of a CMS Plan of Correction in resolving identified deficiencies in long-term care facilities. Your evaluation should be based on the specific details of the deficiency and the corresponding Plan of Correction provided. Follow these guidelines:

1. Input Structure: You will receive text inputs including `deficiency_text`, a detailed narrative describing the observed deficiency, including specific incidents and regulatory references; `plan_of_correction_text`, the facility’s proposed corrective actions to address the deficiency; `ftag`, the specific regulatory tag associated with the deficiency; and `scope_severity`, the severity and scope level of the deficiency, indicating its impact on residents.

2. Evaluation Criteria: Assess whether the Plan of Correction provides clear, actionable steps to correct the deficiency. Determine if the Plan includes specific actions addressing resident-specific corrections, systemic changes, staff education, and ongoing monitoring. Identify if the Plan outlines a timeline for implementation and a mechanism for quality assurance or follow-up. Look for any vague or boilerplate language that may undermine the effectiveness of the Plan.

3. Response Types: Classify the Plan of Correction as adequate if it sufficiently addresses the identified deficiencies with clear, actionable solutions and monitoring mechanisms; not adequate if it fails to address the core issues outlined in the deficiency, lacks specificity, or does not include necessary follow-up actions; and partial if it includes some corrective actions but lacks comprehensive strategies or fails to address major components of the deficiency.

4. Justification: Provide a clear and concise rationale for your classification, referencing specific elements from both the deficiency text and the Plan of Correction. Highlight any strengths and weaknesses in the Plan based on the criteria outlined above.

5. Feedback Utilization: Incorporate feedback from previous evaluations to refine your assessment process, ensuring that you identify material omissions and weaknesses tied directly to the deficiency narrative rather than relying solely on generalizable strategies.

C LLM-Assisted TraceSpec Constructor Outputs

The adaptive construction module asks an LLM constructor to propose a domain-specific TRACE-

SPEC from a few examples, then passes the proposal through the executable utility gate. Below are the complete substantive outputs for the three domains. Full JSON artifacts, including token metadata, are released with the code under [outputs/adaptive_trace_framework/llm_schema_proposals/](https://github.com/LLM-SEC/outputs/adaptive_trace_framework/llm_schema_proposals/).

C.1 SEC Constructor Output

Domain: sec.
Task name: visible-evidence response-resolution review.
Decision: go-primary-benchmark.
Role mapping: request=sec_comment; response=company_response; evidence=retrieved_snippets, amended_evidence_best_snippet; label=visible_evidence_resolution_label; feedback=scalar_feedback, category_feedback, full_feedback, feedback_sec_request, feedback_company_action, feedback_evidence_summary, feedback_unmet_requirement; external_corroboration=real_followup_text, has_real_followup; meta-data=review_thread_id, cik, review_file_no, issue_category, response_year.
Label policy: resolved iff the visible amended-filing evidence substantially satisfies the material SEC request; unresolved iff a material requested item, quantification, exhibit, document, legal/accounting analysis, or other obligation is still missing from the visible evidence.
Feedback policy: scalar=binary correctness against the visible-evidence label; category=issue category plus coarse gap/category feedback; full text=natural-language feedback describing the SEC request, company action, visible evidence coverage, and unmet requirement; external text=later same-topic SEC follow-up text, used only as noisy corroboration. Forbidden at test time: feedback fields, full_feedback, amended-evidence adjudication fields, and real_followup_text.
Utility scores: role observability 5.0; evidence availability 4.5; label viability 4.0; feedback richness 5.0; external corroboration 4.0; release viability 4.5; optimization suitability 5.0; mean 4.571.
Risks: visible-evidence labels are not expert legal/accounting review; retrieval-conditioned labels can be affected by missing amended-filing evidence.
Minimal experiment: review a small subset to validate role mapping and feedback richness.
Paper claim: the role mapping and feedback policy align with SEC review because company responses can be evaluated against visible amended-filing evidence and later follow-up.

C.2 CMS Constructor Output

Domain: cms.
Task name: deficiency-to-plan adequacy review.
Decision: go-cross-regulatory-adaptation.
Role mapping: request=deficiency_text;

response=plan_of_correction_text;
 evidence=plan_of_correction_text,
 cms_poc_covered_elements,
 cms_poc_missing_elements; la-
 bel=cms_poc_binary_label,
 cms_poc_adequacy_label; feed-
 back=cms_poc_optimizer_feedback,
 cms_poc_missing_elements,
 cms_poc_covered_elements,
 cms_poc_reason; external corrob-
 oration=deficiency_corrected,
 correction_date, days_to_correction;
 metadata=source_document_id, ccn,
 ftag, ftag_group, scope_severity,
 severity_band, state, report_year.
Label policy: adequate iff the Plan of Correction
 materially covers resident-specific harm, affected
 population, systemic prevention, monitoring,
 responsible party, and credible completion; not
 adequate iff material remediation elements are
 missing or only generic.
Feedback policy: scalar=binary correctness
 against cms_poc_binary_label; cate-
 gory=ftag_group, severity band, and adequacy
 subtype; full text=covered elements, missing
 elements, and adjudicated adequacy ratio-
 nale; external text=CMS correction metadata,
 treated as weak outcome metadata rather than
 gold adequacy. Forbidden at test time: all
 label/rationale/feedback fields and correction
 outcome metadata.
Utility scores: role observability 4.5; evidence
 availability 3.5; label viability 4.0; feedback
 richness 4.5; external corroboration 2.5; release
 viability 4.0; optimization suitability 4.0; mean
 3.857.
Risks: correction-date metadata is too skewed to
 serve as the main adequacy label; the plan itself is
 both response and corrective artifact, so evidence
 is less separable than in SEC amended filings.
Minimal experiment: 500 examples; labels:
 367 not adequate and 133 adequate; category
 distribution includes resident care/safety,
 other, abuse/neglect, resident rights, care-
 planning/records, infection control, and food
 safety.
Paper claim: the framework adapts the CMS
 deficiency-to-plan review process by inducing
 a domain-specific adequacy rubric rather than
 transferring the SEC checklist verbatim.

1418 C.3 FDA Constructor Output

1419 **Domain:** fda.
1420 **Task name:** warning-to-closeout resolution trac-
1421 ing.
1422 **Decision:** reshape-before-benchmark.
1423 **Role mapping:** request=warning_excerpt; re-
1424 sponse=none; evidence=closeout_excerpt; la-
1425 bel=has_closeout, days_to_closeout; feed-
1426 back=closeout_excerpt; external corrobor-
1427 ation=closeout_issue_date, closeout_url;
1428 metadata=product, subject, issuing_office,
1429 warning_issue_date, warning_url.
1430 **Label policy:** the current public snapshot sup-
1431 ports resolution tracing from warning letter to
1432 closeout letter. It does not support SEC-style
1433 response-adequacy review because public com-
1434 pany response letters are almost absent.
1435 **Feedback policy:** scalar=closeout existence or
1436 delay bucket after adding negative/non-closeout

controls; category=product, subject, issuing of-	1437
office, and violation family; full text=closeout lan-	1438
guage describing FDA's evaluation of corrective	1439
actions; external text=closeout letter text. Forbid-	1440
den at test time for pre-closeout prediction tasks:	1441
closeout_excerpt.	1442
Utility scores: role observability 3.0; evidence	1443
availability 2.0; label viability 1.0; feedback rich-	1444
ness 3.0; external corroboration 4.0; release viabil-	1445
ity 4.0; optimization suitability 2.5; mean 2.786.	1446
Risks: public company response letters are al-	1447
most absent, so request-response-evidence review	1448
is not directly observable; the linked candidate	1449
set is positive-only unless negative or not-yet-	1450
closeout controls are added.	1451
Minimal experiment: use FDA as an adaptive-	1452
construction probe; build a benchmark only after	1453
adding non-closeout controls or contrastive nega-	1454
tives.	1455
Paper claim: FDA provides public warning and	1456
closeout letters with regulatory resolution lan-	1457
guage; it is a useful stress test because a good	1458
construction agent should reshape the task instead	1459
of forcing an SEC-style schema.	1460

D FDA Prompt Calibration and Example

FDA was not optimized with GEPA in this version. Instead, it illustrates the recursive construction-and-calibration loop for a third domain. The constructor first reshapes the corpus into warning-to-closeout trace verification. The strict reviewer then overrejects true closeouts, so we calibrate the domain rubric while keeping the same hidden-field discipline.

D.1 FDA Generic Prompt 1470

You are given an FDA warning letter and a candidate FDA closeout letter. Decide whether the closeout letter corresponds to and resolves the warning letter. Return matched if they are the same regulatory trace, otherwise mismatched.

D.2 FDA Strict RegTrace Prompt 1476

You are a RegTrace reviewer for FDA warning-to-closeout traces. Use only the visible warning letter, candidate closeout letter, and high-level FDA metadata. Do not assume that a closeout letter is correct merely because it uses similar FDA boilerplate or the same product area. Check trace alignment in this order: firm or recipient identity, CMS/reference number or warning date, product/violation family, issuing office, and whether the closeout says FDA evaluated corrective actions for the same warning. Return matched only when the candidate closeout is visibly the resolution evidence for this warning. Return mismatched when it appears to be a different firm, different warning, different reference number, or merely a similar regulatory template.

D.3 FDA Calibrated RegTrace Prompt 1493

You are a RegTrace reviewer for FDA warning-
to-closeout traces. Use only the visible warning
letter, candidate closeout letter, and high-level

1497 FDA metadata. Return matched when the can-
 1498 didate closeout is visibly the resolution letter for
 1499 the same FDA warning trace. Strong matching
 1500 signals include the same firm or recipient, same
 1501 CMS/reference number, explicit mention of the
 1502 same warning-letter date, same product or viola-
 1503 tion family, and closeout language saying FDA
 1504 evaluated corrective actions in response to that
 1505 warning. Do not require the closeout letter to re-
 1506 peat every violation from the warning letter; FDA
 1507 closeout letters are often short boilerplate. Return
 1508 mismatched when the candidate closeout appears
 1509 to concern a different firm, different warning date,
 1510 different reference/CMS number, or only shares a
 1511 broad product area.

1512 **D.4 FDA Example: Calibration Repairs** 1513 **Over-Rejection**

1514 One true matched pair involved H2 Beverages, Inc.
 1515 The warning letter was dated June 14, 2022, and
 1516 the candidate closeout letter was dated March 22,
 1517 2023. The strict reviewer rejected the pair because
 1518 the closeout did not repeat enough of the warning-
 1519 letter substance and appeared to use different prod-
 1520 uct wording. The calibrated reviewer predicted
 1521 matched, explaining that the firm identity matched,
 1522 the warning date matched, the product area aligned,
 1523 and the closeout explicitly stated that FDA eval-
 1524 uated corrective actions in response to the same
 1525 warning. This is the FDA analogue of the CMS
 1526 calibration story: a domain reviewer should verify
 1527 trace identity, but it should not over-require a short
 1528 closeout letter to restate every violation.

1529 **E Additional Dataset Details**

1530 The most common issue categories are other disclo-
 1531 sure issues (183), risk factors (91), SPAC disclosure
 1532 (78), revenue recognition (75), crypto (28), non-
 1533 GAAP (16), and cybersecurity (1). The most com-
 1534 mon gap types are resolved visible evidence (153),
 1535 missing specific disclosure (140), missing quantifi-
 1536 cation (59), resolved-but-regulator-requested-more-
 1537 detail (38), missing exhibit or document (29), in-
 1538 complete accounting analysis (25), incomplete le-
 1539 gal or regulatory analysis (25), and other visible
 1540 evidence gap (3).

1541 **F Additional Qualitative Cases**

1542 The qualitative appendix used during analysis in-
 1543 cludes full SEC comments, company responses,
 1544 amended evidence snippets, method predictions,
 1545 GEPA-full rationales, and later SEC follow-up text
 1546 for selected cases. These include stablecoin ter-
 1547 minology, financial forecast assumptions, PIPE
 1548 price-per-share disclosure, Item 404 related-party

disclosure, expense breakout and trend discussion,
 and SPAC transaction-background disclosure. The
 CMS analysis artifacts similarly include deficiency
 narratives, Plans of Correction, covered and miss-
 ing elements, optimizer feedback, and predictions
 from generic, scalar, category, and full-feedback
 reviewers. The two representative CMS cases in
 Section 9 illustrate the two useful directions of
 feedback learning: accepting a materially adequate
 POC despite generic over-skepticism, and rejecting
 a plausible POC that addresses only one component
 of a multi-part deficiency.

1560